Elevated Foreshock Productivity and Two-Phase Centroid Migration Preceding the 2025 M_w 8.8 Kamchatka Earthquake

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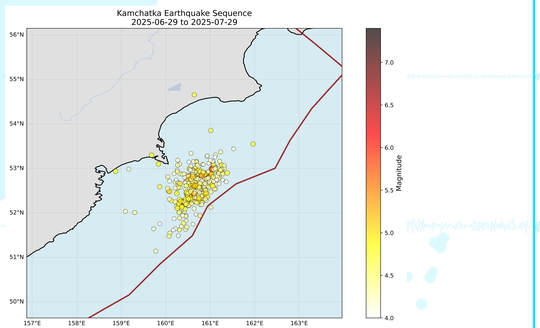
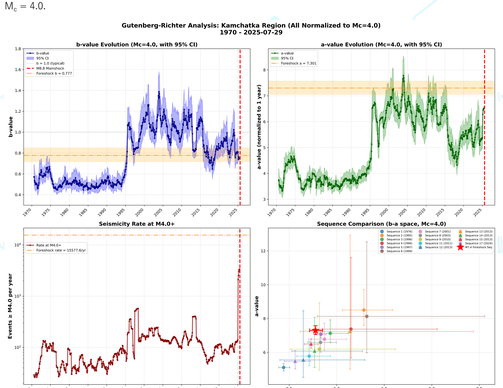
Sequence Background

The July 29, 2025, M_w 8.8 Kamchatka Peninsula earthquake was preceded by a well-defined 10-day foreshock sequence, including one M_w 7.4 and three M_w 6.6 events, that began on July 20. The Kuril-Kamchatka trench is a seismically productive region with a history of large earthquakes, including a M9.0 event in 1952, and recently a M_w 8.8 in 2025. This study presents a statistical and spatiotemporal characterization of the foreshock sequence preceding the 2025 M_w 8.8 event relative to regional background seismicity and 17 other high-activity sequences identified in the 1970-2025 IRIS FDSN catalog. These results are intended to serve as a quantitative baseline for future analysis.

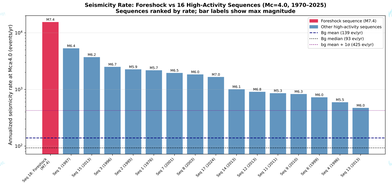


Earthquake catalog data was obtained from the IRIS FDSN web service for the period of 1970-2025, within 350 km of the M_w 8.8 epicenter (51.89° N, 159.76° E), with a minimum magnitude of M 2.5. All analyses were normalized to a fixed magnitude of completeness M_c = 4.0. The observed M_c (MAXC method) averages ~4.6 before 1995 and ~3.8 post-1995. For consistency, statistical comparisons are restricted to the post-1995 period as the primary reference. Pre-1995 windows are shown for context but are subject to bias in b-value, a-value, and seismicity rate due to catalog incompleteness below M_c = 4.0.



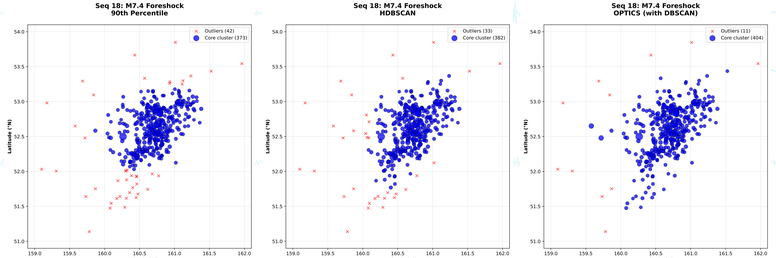
Gutenberg-Richter parameters were estimated using the Aki-Utsu maximum likelihood method with bootstrap uncertainty quantification (1,000–5,000 iterations). High-activity sequences were identified using a sliding-window seismicity-rate threshold (3× the median background rate, minimum 5-day duration, minimum 25 events).

Sequence Background (cont.)



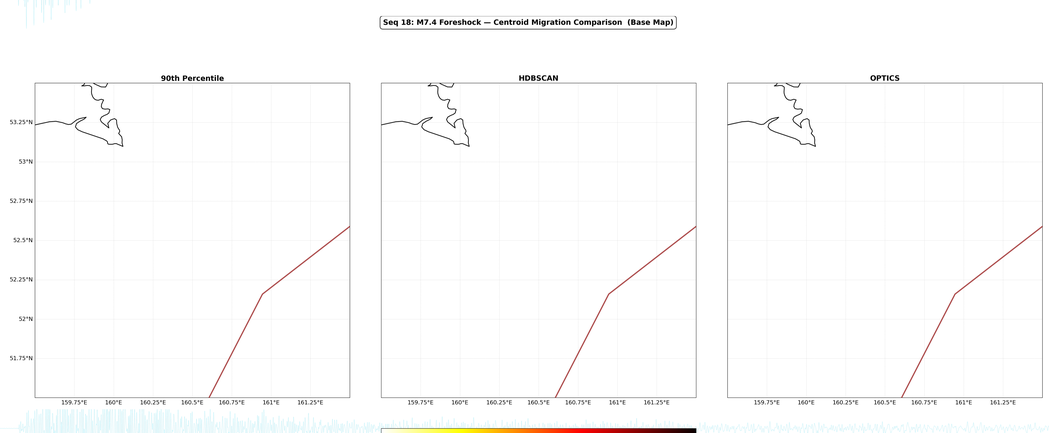
The foreshock sequence yielded b = 0.78 ± 0.07, placing it at approximately the 10th percentile (-1.29 σ) of the post-1995 background distribution (mean 0.97 ± 0.15, n = 380 windows). The foreshock b-value falls within the background range but in the lower tail. No significant b-value anomaly is identified. The a-value of 7.30 ± 0.27 is +1.86 σ above the post-1995 background mean (6.02 ± 0.69), indicating elevated seismicity productivity relative to the completeness-stable catalog period. For the full catalog, the foreshock b value is +0.04 σ from the background mean of 0.78 ± 0.25, and the foreshock a value is +1.84 σ from the background mean of 5.04 ± 1.23. The annualized foreshock seismicity rate of 15,577 M≥4.0 events/year exceeded all 664 background windows in the 55-year catalog, ranking above the >99.99th percentile in both the full catalog and post-1995 period. Among the 18 identified high-activity sequences, the foreshock sequence ranked first in seismicity rate at all tested M_c thresholds (3.5 - 5.0). The +54 σ annualized seismicity rate anomaly reported relative to the background assumes a normal distribution. The background rate distribution is strongly right-skewed (median 92 events/year vs mean 114, from pre-foreshock sliding windows), making the percentile rank the more appropriate metric.

Spatial Sequence Classification and Centroid Migration Analysis

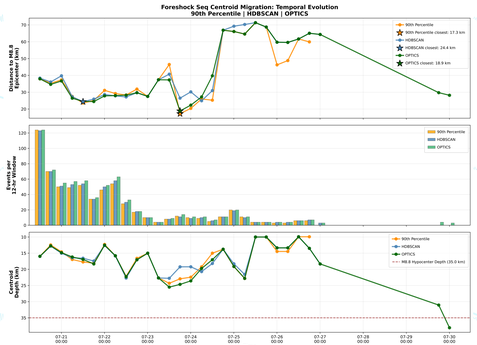


Centroid migration was computed using 12-hour sliding windows with a 6-hour step size and a minimum of 3 events per window. Spatial sequence event classification was applied using three independent methods: 90th percentile distance threshold, HDBSCAN (min_cluster_size = 3, min_samples = 1, ε = 0.3°), and OPTICS (DBSCAN extraction, min_samples = 2, ε = 0.3°). Centroids are spatial means unweighted by magnitude. A filter was applied to the HDBSCAN and OPTICS clusters to include events within 30km of the central cluster.

Centroid Migration Across All Spatial Classification Methods (This is an interactive flipbook!)



Centroid Migration Analysis (cont.)



Centroid migration analysis (12-hr sliding windows, 6-hr step, minimum 3 events/window) reveals a two-phase spatial structure. During the initial foreshock phase (July 20–24), the seismicity centroid exhibited loosely constrained, variable net displacement towards the M_w 8.8 epicenter, reaching closest approach of 17.3-24.4 km (range across all spatial event classification methods) on July 23 (July 21 on 90th percentile), concurrent with significant decay in seismicity rate. The sequence originated ~38km from the mainshock epicenter. On July 24, a M_w 6.0 event nucleated a spatially distinct sub-cluster ~21km to the northeast of the original M_w 7.4 cluster, producing an abrupt centroid displacement of 27.1-41.6 km (range across all methods). Activity in the SW cluster terminated following the M_w 6.0. The centroid remained anchored in the NE sub-cluster, at ~66km from the M_w 8.8 hypocenter through the end of the foreshock sequence. This two-phase structure is robust across all three spatial event classification methods. A late re-approach visible in the OPTICS-cleaned catalog is not interpreted due to low event counts per-window.

Conclusions & Future Work

→ The M_w 7.4 2025 Kamchatka foreshock sequence had the highest seismicity rate among all high-activity sequences in the 55-year regional catalog at every tested magnitude threshold, ranking above the >99.99th percentile of the background activity. The a-value was moderately elevated from the background mean (+1.86 σ, post-1995 reference).

→ The foreshock sequence b-value 0.78 ± 0.07 falls within the background range but in the lower tail (-1.29 σ) of the post-1995 distribution. No statistically significant anomaly is identified from the b-value alone.

→ Centroid migration analysis reveals sequential activation of two spatial clusters: an initial cluster exhibiting loosely constrained variable drift toward the M_w 8.8 hypocenter before significant activity decay, and a NE sub-cluster nucleated by a M_w 6.0 on Jul 24, which was spatially persistent through its decay. The terminal foreshock centroid was displaced further from the eventual M_w 8.8 nucleation point than at sequence onset.

→ These results provide a quantitative spatiotemporal characterization of the foreshock sequence to be utilized as a baseline in future studies.

Future Work:
→ GS RAS regional catalog (M<3.5) will be requested. Acquisition will enable extended analysis and potentially refined spatial clustering.
→ Formal directional consistency metrics for centroid migration.
→ Comparison with documented foreshock migration on similar subduction zones.
→ Formal directional consistency metrics for centroid migration.

This is a preliminary analysis. All code used in this analysis (and in future continuations) will be uploaded to my public GitHub repository at:
https://github.com/seismogenesis/kamchatka-earthquake

Questions? Contact me:
My LinkedIn:
My GitHub repo:

Acknowledgements:
IRIS DMC (2025), IRIS FDSN Web Services, https://service.iris.edu/
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